21. The conjugates of $\sqrt[3]{2}$ over $\mathbb{Q}(i\sqrt{3})$ are

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\frac{-1+i\sqrt{3}}{2}, \quad \text{and} \quad \sqrt[3]{2}\frac{-1-i\sqrt{3}}{2}.$$

Maps of $\sqrt[3]{2}$ into each of them give rise to the only three automorphisms in $G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}(i\sqrt{3}))$. Let σ be the automorphism such that $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\frac{-1+i\sqrt{3}}{2}$. Then σ must be a generator of this group of order 3, because σ is not the identity map, and every group of order 3 is cyclic. Thus the automorphism group is isomorphic to $\mathbb{Z}_3$.

22. **a.** Each automorphism of E leaving F fixed is a one-to-one map that carries each zero of $f(x)$ into one of its conjugates, which must be a zero of an irreducible factor of $f(x)$ and hence is also a zero of $f(x)$. Thus each automorphism gives rise to a one-to-one map of the set of zeros of $f(x)$ onto itself, that is, it acts as a permutation on the zeros of $f(x)$.

b. Because E is the splitting field of $f(x)$ over F, we know that $E = F(\alpha_1, \alpha_2, \cdots, \alpha_n)$ where $\alpha_1, \alpha_2, \cdots, \alpha_n$ are the zeros of $f(x)$. As Exercise 33 of Section 48 shows, an automorphism σ of E leaving F fixed is completely determined by the values $\sigma(\alpha_1), \sigma(\alpha_2), \cdots, \sigma(\alpha_n)$ that is, by the permutation of the zeros of $f(x)$ given by σ.

c. We associate with each $\sigma \in G(E/F)$ its permutation of the zeros of $f(x)$ in E. Part(**b**) shows that different elements of $G(E/F)$ produce different permutations of the zeros of $f(x)$. Because multiplication $\sigma\tau$ in $G(E/F)$ is function composition and because multiplication of the permutations of zeros is again composition of these same functions, with domain restricted to the zeros of $f(x)$, we see that $G(E/F)$ is isomorphic to a subgroup of the group of all permutations of the zeros of $f(x)$.

23. **a.** We have $|G(E/\mathbb{Q})| = 2 \cdot 3 = 6$, because $\{\mathbb{Q}(i\sqrt{3}) : \mathbb{Q}\} = 2$ since $\text{irr}(i\sqrt{3}, \mathbb{Q}) = x^2 + 3$ and $\{\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(i\sqrt{3})\} = 3$ because $\text{irr}(\sqrt[3]{2}, \mathbb{Q}(i\sqrt{3})) = x^3 - 2$. The index is multiplicative by Corollary 49.10.

b. Because E is the splitting field of $x^3 - 2$ over $\mathbb{Q}$, Exercise 22 shows that $G(E/\mathbb{Q})$ is isomorphic to a subgroup of the group of all permutations of the three zeros of $x^3 - 2$ in E. Because the group of all permutations of three objects has order 6 and $|G(E/\mathbb{Q})| = 6$ by Part(**a**), we see that $G(E/\mathbb{Q})$ is isomorphic to the full symmetric group on three letters, that is, to S_3.

24. We have $x^p = (x-1)(x^{p-1} + \cdots + x + 1)$, and Corollary 23.17 shows that the second of these factors, the cyclotomic polynomial $\Phi_p(x)$, is irreducible over the field $\mathbb{Q}$. Let ζ be a zero of $\Phi_p(x)$ in its splitting field over $\mathbb{Q}$. Exercise 36a of Section 48 shows that then $\zeta, \zeta^2, \zeta^3, \cdots, \zeta^{p-1}$ are distinct and are all zeros of $\Phi_p(x)$. Thus all zeros of $\Phi_p(x)$ lie in the simple extension $\mathbb{Q}(\zeta)$, so $\mathbb{Q}(\zeta)$ is the splitting field of $x^p - 1$ and of course has degree $p - 1$ over $\mathbb{Q}$ because $\Phi_p(x) = \text{irr}(\zeta, \mathbb{Q})$ has degree $p - 1$.

25. By Corollary 49.5, there exists an isomorphism $\phi : \overline{F} \to \overline{F'}$ leaving each element of F fixed. Because the coefficients of $f(x) \in F[x]$ are all left fixed by ϕ, we see that ϕ carries each zero of $f(x)$ in $\overline{F}$ into a zero of $f(x)$ in $\overline{F'}$. Because the zeros of $f(x)$ in $\overline{F}$ generate its splitting field E in $\overline{F}$, we see that $\phi[E]$ is contained in the splitting field E' of $f(x)$ in $\overline{F'}$. But the same argument can be made for ϕ^{-1}; we must have $\phi^{-1}[E'] \subseteq E$. Thus ϕ maps E onto E', so these two splitting fields of $f(x)$ are isomorphic.

51. Separable Extensions

1. Because $\sqrt[3]{2}\sqrt{2} = 2^{1/3}2^{1/2} = 2^{5/6}$, we have $\sqrt[6]{2} = 2/(\sqrt[3]{2}\sqrt{2})$ so $\mathbb{Q}(\sqrt[6]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \sqrt{2})$. Because $\sqrt[3]{2} = (\sqrt[6]{2})^2$ and $\sqrt{2} = (\sqrt[6]{2})^3$, we have $\mathbb{Q}(\sqrt[3]{2}, \sqrt{2}) \subseteq \mathbb{Q}(\sqrt[6]{2})$, so $\mathbb{Q}(\sqrt[6]{2}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{2})$. We can take $\alpha = \sqrt[6]{2}$.

2. Because $(\sqrt[4]{2})^3(\sqrt[6]{2}) = 2^{3/4}2^{1/6} = 2^{9/12}2^{2/12} = 2^{11/12}$, we see that $\sqrt[12]{2} = 2/[(\sqrt[4]{2})^3(\sqrt[6]{2})]$ so $\mathbb{Q}(\sqrt[12]{2}) \subseteq \mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2})$. Because $\sqrt[4]{2} = (\sqrt[12]{2})^3$ and $\sqrt[6]{2} = (\sqrt[12]{2})^2$, we have $\mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2}) \subseteq \mathbb{Q}(\sqrt[12]{2})$, so $\mathbb{Q}(\sqrt[12]{2}) = \mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2})$. We can take $\alpha = \sqrt[12]{2}$.

3. We try $\alpha = \sqrt{2} + \sqrt{3}$. Squaring and cubing, we find that $\alpha^2 = 5 + 2\sqrt{2}\sqrt{3}$ and $\alpha^3 = 11\sqrt{2} + 9\sqrt{3}$. Because

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{2} \text{ and } \sqrt{3} = \frac{11\alpha - \alpha^3}{2},$$

we see that $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

4. Of course $\mathbb{Q}(i\sqrt[3]{2}) \subseteq \mathbb{Q}(i, \sqrt[3]{2})$. Because $i = -(i\sqrt[3]{2})^3/2$ and $\sqrt[3]{2} = -2/(i\sqrt[3]{2})^2$, we see that $\mathbb{Q}(i, \sqrt[3]{2}) \subseteq \mathbb{Q}(i\sqrt[3]{2})$. Thus $\mathbb{Q}(i, \sqrt[3]{2}) = \mathbb{Q}(i\sqrt[3]{2})$, so we can take $\alpha = i\sqrt[3]{2}$.

5. The definition is incorrect. Replace $F[x]$ by $\overline{F}[x]$ at the end.

Let $\overline{F}$ be an algebraic closure of a field F. The **multiplicity of a zero** $\alpha \in \overline{F}$ of a polynomial $f(x) \in F[x]$ is $\nu \in \mathbb{Z}^+$ if and only if $(x - \alpha)^\nu$ is the highest power of $x - \alpha$ that is a factor of $f(x)$ in $\overline{F}[x]$.

6. The definition is correct.

7. (See the answer in the text.)

8. F T T F F T T T T T

9. We are given that α is separable over F, so by definition, $F(\alpha)$ is a separable extension over F. Because β is separable over F, it follows that β is separable over $F(\alpha)$ because $q(x) = \text{irr}(\beta, F(\alpha))$ divides $\text{irr}(\beta, F)$ so β is a zero of $q(x)$ of multiplicity 1. Therefore $F(\alpha, \beta)$ is a separable extension of F by Theorem 51.9. Corollary 51.10 then asserts that each element of $F(\alpha, \beta)$ is separable over F. In particular, $\alpha \pm \beta, \alpha\beta$, and α/β if $\beta \neq 0$ are all separable over F.

10. We know that $[\mathbb{Z}_p(y) : \mathbb{Z}_p(y^p)]$ is at most p. If we can show that $\{1, y, y^2, \cdots, y^{p-1}\}$ is an independent set over $\mathbb{Z}_p(y^p)$, then by Theorem 30.19, this set could be enlarged to a basis for $\mathbb{Z}_p(y)$ over $\mathbb{Z}_p(y^p)$. But because a basis can have at most p elements, it would already be a basis, and $[\mathbb{Z}_p(y) : \mathbb{Z}_p(y^p)] = p$, showing that $\text{irr}(y, \mathbb{Z}_p(y^p))$ would have degree p and must therefore be $x^p - y^p$. Thus our problem is reduced to showing that $S = \{1, y, y^2, \cdots, y^{p-1}\}$ is an independent set over $\mathbb{Z}_p(y^p)$.

Suppose that

$$\frac{r_0(y^p)}{s_0(y^p)} \cdot 1 + \frac{r_1(y^p)}{s_1(y^p)} \cdot y + \frac{r_2(y^p)}{s_2(y^p)} \cdot y^2 + \cdots + \frac{r_{p-1}(y^p)}{s_{p-1}(y^p)} \cdot y^{p-1} = 0$$

where $r_i(y^p), s_i(y^p) \in \mathbb{Z}_p[y^p]$ for $i = 0, 1, 2, \cdots, p-1$. We want to show that all these coefficients in $\mathbb{Z}_p(y^p)$ must be zero. Clearing denominators, we see that it is no loss of generality to assume that all $s_i(y^p) = 1$ for $i = 0, 1, 2, \cdots, p-1$. Now the powers of y appearing in $r_i(y^p)(y^i)$ are all congruent to i modulo p, and consequently no terms in this expression can be combined with any terms of $r_j(y^p)(y^j)$ for $j \neq i$. Because y is an indeterminant, we then see that this linear combination of elements in S can be zero only if all the coefficients $r_i(y^p)$ are zero, so S is an independent set over $\mathbb{Z}_p(y^p)$, and we are done.

11. Let E be an algebraic extension of a perfect field F and let K be a finite extension of E. To show that E is perfect, we must show that K is a separable extension of E. Let α be an element of K. Because $[K : E]$ is finite, α is algebraic over E. Because E is algebraic over F, then α is algebraic over F by

Exercise 31 of Section 31. Because F is perfect, α is a zero of $\text{irr}(\alpha, F)$ of multiplicity 1. Because $\text{irr}(\alpha, E)$ divides $\text{irr}(\alpha, F)$, we see that α is a zero of $\text{irr}(\alpha, E)$ of multiplicity 1, so α is separable over E by the italicized remark preceding Theorem 51.9. Thus each $\alpha \in K$ is separable over E, so K is separable over E by Corollary 51.10.

12. Because K is algebraic over E and E is algebraic over F, we have K algebraic over F by Exercise 31 of Section 31. Let $\beta \in K$ and let $\beta_0, \beta_1, \cdots, , \beta_n$ be the coefficients in E of $\text{irr}(\beta, E)$. Because β is a zero of $\text{irr}(\beta, E)$ of algebraic multiplicity 1, we see that $F(\beta_0, \beta_1, \cdots, \beta_n, \beta)$ is a separable extension of $F(\beta_0, \beta_1, \cdots, \beta_n)$, which in turn is a separable extension of F by Corollary 51.10. Thus we are back to a tower of finite extensions, and deduce from Theorem 51.9 that $F(\beta_0, \beta_1, \cdots, \beta_n, \beta)$ is a separable extension of F. In particular, β is separable over F. This shows that every element of K is separable over F, so by definition, K is separable over F.

13. Exercise 9 shows that the set S of all elements in E that are separable over F is closed under addition, multiplication, and division by nonzero elements. Of course 0 and 1 are separable over F, so Exercise 9 further shows that S contains additive inverses and reciprocals of nonzero elements. Therefore S is a subfield of E.

14. **a.** We know that the nonzero elements of E form a cyclic group E^* of order $p^n - 1$ under multiplication, so all elements of E are zeros of $x^{p^n} - x$. (See Section 33.) Thus for $\alpha \in E$, we have

$$\begin{aligned} {\sigma_p}^n(\alpha) &= {\sigma_p}^{n-1}(\sigma_p(\alpha)) = {\sigma_p}^{n-1}(\alpha^p) = {\sigma_p}^{n-2}(\sigma_p(\alpha^p)) \\ &= {\sigma_p}^{n-2}(\sigma_p(\alpha))^p = {\sigma_p}^{n-2}((\alpha^p)^p) = {\sigma_p}^{n-2}(\alpha^{p^2}) \\ &= \cdots = \alpha^{p^n} = \alpha \end{aligned}$$

so ${\sigma_p}^n$ is the identity automorphism. If α is a generator of the group E^*, then $\alpha^{p^i} \neq \alpha$ for $i < n$, so we see that n is indeed the order of σ_p.

b. Section 33 shows that E is an extension of $\mathbb{Z}_p$ of order n, and is the splitting field of any irreducible polynomial of degree n in $\mathbb{Z}[x]$. Because E is a separable extension of the finite perfect field $\mathbb{Z}_p$, we see that $|G(E/F)| = \{E : F\} = [E : F] = n$. Since $\sigma_p \in G(E/F)$ has order n, we see $G(E/F)$ is cyclic of order n.

15. **a.** Let $f(x) = \sum_{i=0}^{\infty} a_i x^i$ and $g(x) = \sum_{i=0}^{\infty} b_i x^i$. Then

$$\begin{aligned} D(f(x) + g(x)) &= D\left(\sum_{i=0}^{\infty}(a_i + b_i)x^i\right) \\ &= \sum_{i=1}^{\infty}(i \cdot 1)(a_i + b_i)x^{i-1} \\ &= \sum_{i=1}^{\infty}(i \cdot 1)a_i x^{i-1} + \sum_{i=1}^{\infty}(i \cdot 1)b_i x^{i-1} \\ &= D(f(x)) + D(g(x)). \end{aligned}$$

thus D is a homomorphism of $\langle F[x], + \rangle$.

b. If F has characteristic zero, then $\text{Ker}(D) = F$.

c. If F has characteristic p, then $\text{Ker}(D) = F[x^p]$.

16. a. Let $f(x) = \sum_{i=0}^{\infty} a_i x^i$. Then

$$\begin{aligned} D(af(x)) &= D\left(\sum_{i=0}^{\infty} aa_i x^i\right) = \sum_{i=1}^{\infty}(i\cdot 1)aa_i x^{i-1} \\ &= a\sum_{i=1}^{\infty}(i\cdot 1)a_i x^{i-1} = aD(f(x)). \end{aligned}$$

b. We use induction on $n = \deg(f(x)g(x))$. If $n = 0$, then $f(x), g(x), f(x)g(x) \in F$ and $D(f(x)) = D(g(x)) = D(f(x)g(x)) = 0$ by Part(**b**) and Part(**c**) of Exercise 15. Suppose the formula is true for $n < k$, and let us prove it for $n = k < 0$. Write $f(x) = h(x) + a_r x^r$ where $a_r x^r$ is the term of highest degree in $f(x)$. Similarly write $g(x)$ as $g(x) = q(x) + b_s x^s$. Then

$$f(x)g(x) = h(x)q(x) + h(x)b_s x^s + a_r x^r q(x) + a_r b_s x^{r+s}.$$

Of these four terms, all are of degree less than $k = r + s$ except for the last term, so by Part(**a**) of Exercise 15 and our induction hypothesis, we have

$$\begin{aligned} D(f(x)g(x)) &= h(x)q'(x) + h'(x)q(x) \\ &\quad + h(x)(s\cdot 1)b_s x^{s-1} + h'(x)b_s x^s \\ &\quad + a_r x^r q'(x) + (r\cdot 1)a_r x^{r-1}q(x) \\ &\quad + [(r+s)\cdot 1]a_r b_s x^{r+s-1}. \end{aligned}$$

We notice that we have

$$h(x)q'(x) + h(x)(s\cdot 1)b_s x^{s-1} = h(x)g'(x)$$

and

$$h'(x)q(x) + (r\cdot 1)a_r x^{r-1}q(x) = f'(x)q(x),$$

so we can continue with

$$\begin{aligned} D(f(x)g(x)) &= h(x)g'(x) + f'(x)q(x) + h'(x)b_s x^s + a_r x^r q'(x) \\ &\quad + (s\cdot 1)a_r b_s x^{r+s-1} + (r\cdot 1)a_r b_s x^{r+s-1} \\ &= h(x)g'(x) + f'(x)q(x) \\ &\quad + a_r x^r[q'(x) + (s\cdot 1)b_s x^{s-1}] \\ &\quad + [h'(x) + (r\cdot 1)a_r x^{r-1}]b_s x^s \\ &= h(x)g'(x) + f'(x)q(x) + a_r x^r g'(x) + f'(x)b_s x^s \\ &= [h(x) + a_r x^r]g'(x) + f'(x)[q(x) + b_s x^s] \\ &= f(x)g'(x) + f'(x)g(x). \end{aligned}$$

c. We proceed by induction on m. If $m = 1$, the relation becomes $D((f(x))^1) = ((1\cdot 1)f(x)^0 f'(x)$ which reduces to $D(f(x)) = f'(x)$. Thus the relation holds for $m = 1$. Suppose it is true for $m < k$ where $k > 1$. We show it holds for $m = k$. Using Part(**b**), we obtain

$$\begin{aligned} D(f(x)^k) &= D(f(x)(f(x)^{k-1})) \\ &= f(x)[(k-1)f(x)^{k-2}f'(x)] + f'(x)f(x)^{k-1} \\ &= [f(x)(k-1)f(x)^{k-2} + f(x)^{k-1}]f'(x) \\ &= [(k-1)f(x)^{k-1} + f(x)^{k-1}]f'(x) \\ &= kf(x)^{k-1}f'(x) \end{aligned}$$

which completes our induction proof.